Chronic Kidney Disease Prediction using Hybrid CNN-KNN Model

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***Abstract*—**
***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

This section introduces the topic, states the main research question, and sets out the purpose and scope of the study. Chronic Kidney Disease (CKD) exists as a global health problem marked by slow and permanent loss of kidney function over time. Because early CKD presents few or no symptoms, many cases have been identified at a later stage, increasing the risk of complications such as cardiovascular disease, anemia, electrolyte imbalance, and progression to end-stage renal disease that may require dialysis or transplantation . The prevalence of CKD has increased globally in many settings, with marked growth in developing countries, largely related to higher rates of diabetes and hypertension and aging of the population.Traditional diagnostic approaches often depend on many laboratory tests and specialist judgment, which can miss complex, non-linear patterns across clinical measures, especially in the early stages of disease .As electronic health records become more widely available, researchers can apply machine learning and artificial intelligence methods to predict disease based on patient data.Machine learning methods, including logistic regression and k-nearest neighbors (KNN), have been reported to perform well in CKD classification .No single model performs consistently well across all datasets because the data vary between settings. To address this limitation, researchers have proposed ensemble and hybrid methods to improve model stability and performance on new data. In recent work, deep learning models, especially CNNs, have shown strong ability to learn features directly from medical data with limited manual input. Integrating CNN-based feature learning with traditional classifiers can increase accuracy while keeping the model easier to interpret. This study develops and evaluates machine learning, ensemble, and hybrid deep learning models to support early prediction of CKD.This study proposes a hybrid 1D CNN–KNN model in which the CNN extracts features and the KNN performs the final classification. The data are carefully cleaned and prepared, and standard evaluation metrics are used to compare model performance. The dataset has been created and preprocessed with the use of Python tools and techniques before model training taking place. Since the CKD dataset has proned to class imbalance, the SMOTE-Tomek method is being used to balance both CKD and non-CKD examples within the dataset. This method is proving useful in promoting quality learning, which is free from Majority Class Bias and Overlapping examples. After preprocessing, the dataset is being split into a training set and a testing set in a ratio of 80:20. The training set is being used for developing the models, and the testing set was used to evaluate the models. This was to ensure that the models were evaluated in terms of how well they performed in unseen examples. The performance measurement of models is being conducted through regular metrics such as accuracy, precision, recall, F1-score, Cohen's Kappa statistic, and ROC. Bar charts are being used to analyze inter-model comparisons, and ROC curves are being used to analyze sensitivity. A block diagram is being used to represent the whole process for which the mechanism has been designed. This provides a clear idea of all processes conducted from data processing till final evaluation.

II. Literature Survey

Early CKD is often asymptomatic, so diagnosis may occur late, after kidney damage becoming irreversible. Late detection increases the risk of complications, including cardiovascular disease, anaemia, electrolyte imbalance, and progression to end-stage renal disease that may require dialysis or kidney transplantation. Research indicates that early diagnosis plays an important role in slowing disease progression and improving patient survival, especially in settings where access to specialised medical services is limited. Machine learning methods can be applied to early CKD prediction because

electronic health records are now widely available. To analyse clinical parameters and classify patients with CKD, researchers have commonly applied algorithms such as logistic regression, decision trees, naïve Bayes, and KNN. These methods offer clear interpretability and solid accuracy, but the results depend strongly on class balance, the choice of features, and the quality of the data. No single machine learning model performs best in every setting because model performance depends on the specific dataset used. Because deep learning techniques, like CNNs, learn feature representations directly from the data and rely less on manually created features, they frequently achieve strong performance in medical tasks. CNN-based methods have the potential for improving prediction accuracy by representing non-linear relationships between clinical variables. Despite these advantages, deep learning models may not be widely used in clinical practice because they frequently require large datasets and are still challenging to understand. To overcome these difficulties, researchers have developed hybrid models that combine deep learning with conventional machine learning techniques. Hybrid techniques, such as combining a 1D CNN with KNN, using the CNN for extracting features, and then using KNN to classify samples according to similarity in a way that is still understandable. The use of hybrid models in early CKD risk prediction and clinical decision support systems is supported by previous research showing that they frequently offer greater accuracy, better generalization, and clearer interpretability than using machine learning or deep learning alone.

III. Materials and Methodology

The materials and methodology are described in the following subsections. The general architecture of the methodology has been shown in Figure 1 which is concise of Data collection, Data preprocessing, methodology, and Evaluation parameters.

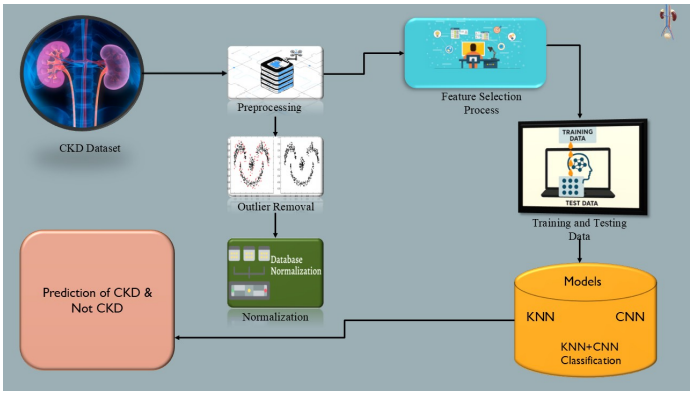


Fig. 1: Hybrid deep learning framework for early detection of CKD.

A. Data Collection

The Kidney Disease Risk Dataset used in this study has been obtained from Kaggle, where it has been openly accessible for academic and research work. Data have been arranged in a tabular structure in which each row corresponds to a single patient and each column has been representing a clinical or laboratory variable associated with kidney health. Mathematically, the data set has been expressed as:

$$D = \{(x_i, y_i)\}_{i=1}^{N} \tag{1}$$

where

$$x_i = [x_{i1}, x_{i2}, \ldots, x_{in}] \in \mathbb{R}^n \tag{2}$$

denotes the feature vector for the i-th patient has been denoting

$$y_i \in \{0, 1\} \tag{3}$$

represents the class label has been indicated whether the patient has been at risk of CKD. The attributes have been included demographic information such as age and gender, clinical measurements like blood pressure, laboratory indicators including serum creatinine and haemoglobin levels, and comorbidity-related variables such as diabetes and hypertension status. In comparison with the hospital dataset reported in earlier work, which consisted of 1,718 samples and 19 features, the present dataset has been varied in both dimensionality and statistical distribution, making direct replication unsuitable and required independent preprocessing and modelling steps.

TABLE I: Dataset Description Used in Methodology

Parameter	Description
Dataset Name	Final Intersection CKD Dataset
Data Source	Processed version of Chronic Kidney Disease clinical dataset
Total Patient Records	400 Instances
Total Selected Features	16 Attributes (15 Clinical Features + 1 Target Variable)
Dataset Type	Mixed (Numerical and Encoded Categorical Variables)
Target Variable	Classification (CKD = 1, Non-CKD = 0)
Data Domain	Medical and Clinical Laboratory Records
Feature Selection Method	Filter, Wrapper, and Embedded Intersection Approach
Purpose of Dataset	Early Detection and Classification of Chronic Kidney Disease
Data Format	Tabular Structured Dataset
Preprocessing Techniques Applied	Missing Value Imputation, Label Encoding, Normalization, Outlier Detection
Dataset Splitting Ratio	Training Data – 70%, Testing Data – 30%
Model Application	Hybrid 1D CNN for Feature Extraction and KNN for Classification

B. Data preprocessing

As commonly observed in medical datasets, incomplete entries and inconsistencies have been present due to unrecorded laboratory investigations, manual input errors, or device-related limitations. To address missing numerical values, statistical imputation has been applied using the mean as shown in equation 4.

$$\bar{x}_j = \frac{1}{N_j} \sum_{i=1}^{N_j} x_{ij} \tag{4}$$

or the median has been required:

$$\tilde{x}_j \tag{5}$$

while categorical variables have been replaced with the most frequent category,

$$Mode(x_j) \tag{6}$$

Outlier detection has been carried out using the interquartile range defined as shown in Equation -7

$$IQR_j = Q_{3j} - Q_{1j} \tag{7}$$

where observations outside the interval has been shown in equation-8

$$[Q_{1j} - 1.5IQR_j,\ Q_{3j} + 1.5IQR_j] \tag{8}$$

are considered extreme. In addition, Z-score analysis has been employed using in equation - 9

$$Z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j} \tag{9}$$

and values satisfying

$$|Z_{ij}| > 3 \tag{10}$$

are flagged as anomalies. Before model training, features have been scaled through min–max normalization which has been shown in equation - 10

$$x'_{ij} = \frac{x_{ij} - x_j^{min}}{x_j^{max} - x_j^{min}} \tag{11}$$

or standardization

$$x''_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j} \tag{12}$$

maintain numerical stability. The prediction task has been formulated as a binary classification problem, where the conditional probability has been modeled as shown in Equation -13

$$P(y = 1 \mid x; \theta) = \sigma(w^T x + b) \tag{13}$$

with the sigmoid function has been shown in equation - 14

$$\sigma(z) = \frac{1}{1 + e^{-z}} \tag{14}$$

Parameter estimation has been performed by minimizing the binary cross-entropy loss, as shown in equation - 15

$$L(\theta) = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \tag{15}$$

and iterative updates followed the gradient descent rule has been shown in equation –16

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_\theta L(\theta) \tag{16}$$

has been denoting the learning rate. This formulation provides a structured and reproducible framework for predicting the risk of CKD.

C. Data Preparation

Data preparation has been a crucial step in developing reliable predictive models. Following the method outlined in the reference study, preprocessing has been considered a necessary phase to improve data quality and confirm that the data set is suitable for analysis. This stage has focused on improving the data structure, fixing missing information, converting categorical variables, and scaling numerical features. A similar strategy has been used for the Kaggle data set in this study.

1) Data Cleaning: Data cleaning has been the next step, beginning with a careful review of the data set to spot issues that could interfere with the analysis. Duplicate records have been removed to prevent repeated information, and obvious inconsistencies have been fixed when possible. The data set has also been checked for unusually large or small values. These values have been evaluated using the Interquartile Range (IQR) method to see if they were outliers that needed attention. This step led to a clearer and more reliable data set.

2) Handling Missing Values: The handling of missing values has been the next crucial step. Missing values have been common in real-world medical data and can weaken the stability of analytical results if not handled. For numerical attributes, replacement values have been chosen using either the mean or the median, depending on the nature of each variable. For categorical attributes, missing entries have been replaced with the most common category. This approach maintained the integrity of the data set while ensuring that it has been completed.

3) Encoding Categorical Variables: The dataset has been included several categorical variables, such as gender, diabetes status, and hypertension. Since machine learning models have worked with numerical data, these attributes have been changed into numeric forms. Binary categories have been encoded using simple label encoding, while variables with multiple categories have been transformed through one-hot encoding. This change allowed the algorithms to use categorical information effectively during training.

4) Feature Scaling: In the final stage of preprocessing, numerical variables have been standardized to bring them onto similar scales. This step has been especially helpful when features vary greatly in size. Z-score normalization has been applied to rescale each variable based on its mean and standard deviation, using the following equation :

$$z = \frac{x - \mu}{\sigma} \tag{17}$$

where x denotes the original feature value, μ is the mean, and σ is the standard deviation.

D. Feature Selection

Selecting the right features has been a key step in preparing the dataset. Instead of using every available variable, the focus has been on identifying which attributes have been genuinely contributing to the prediction task. Keeping unnecessary or repetitive variables has been complicating model training and sometimes reducing performance. By narrowing the dataset to the most meaningful inputs, the analysis becomes easier to interpret, and the model has been able to learn more effectively. Two different approaches guided this process:

1) Univariate Feature Selection: The first approach examined each variable independently. In other words, features has been being analysed one by one to see how much they have been linked to the target outcome. Statistical evaluation has been helping determine whether a particular attribute has been providing useful information. The ANOVA F-test was used

for this purpose, allowing a comparison of how strongly each feature has been relating to the prediction goal. Attributes that showing stronger relationships have been being considered more informative.

2) Recursive Feature Elimination (RFE): To supplement the individual assessment, an iterative selection method has been employed. Recursive Feature Elimination has been works by repeatedly fitting a model and removing the variables that contribute the least. After each training cycle, the weakest features have been being discarded, and the model has been rebuilt using the smaller set. This step-by-step reduction has been continuing until a compact and effective group of variables remains. The resulting subset has been reflects the features that offer the greatest predictive value without adding unnecessary complexity.

E. Model Training and Evaluation

Data Division has been arranging the dataset for learning and assessment before creating the model. In order for the algorithms to find patterns among the variables, the majority of the records has been sent to the training stage. The rest of the data has been being reserved for testing. An 80-20 split has been used, with the smaller portion being used only for assessment and the larger portion being used for training. By maintaining this division, the performance has been being assessed using data that hasn't utilised to train the model. Cross-Checking During training, we used cross-validation to increase the evaluation process's dependability. We repeatedly divided the dataset into smaller pieces using a 10-fold method. One of the datasets has been employed for validation, and the remaining for training in each round. A better approximation of the performance of the model has been achieved by considering the validation data to be rotated over the folds,reducing the risk of the outcome being affected by the configuration of the data. Metrics for Evaluation have been employed to variety of metrics for the evaluation of the outcome of the model. Although precision and recall enabled us to gain deeper insights into the classification patterns, accuracy has been acting as a broad indicator of the correctness of the predictions, while precision and recall have been enabling deeper insights into the classification patterns. The F1-Score has been balancing these two aspects to a certain extent. Sensitivity and specificity have also been employed to gain deeper insights into the manner in which the models have been distinguishing between instances of CKD and Non-CKD.

F. Overall Methodology Workflow

The project has been carried out through multiple steps that working together. The first step has been retrieving the dataset from Kaggle and putting it in the time required to prepare it for analysis. In this step, we performed a number of tasks to clean the data by looking at the records of the dataset and correcting obvious errors, treating missing data, and changing categorical data into numerical format. Some of the numerical data have also been rescaled so that any difference in value would not have an impact on our analysis. Feature selection approaches have been used to determine which features have predictive value when creating the model. Having fewer features to use has been provided better data for training the model. To get a better understanding of what kind of accuracy our results produced from our training and validation sets, we also included cross validation steps into our training and validation process. At the end of the entire process, we evaluated the model using various methods to see the overall performance of the model and obtain relative performance of the models.

IV. Result

A. Analysis of different Features of CKD using Selection

1) Dataset Source and Description: The dataset used in this research was obtained from Kaggle and is publicly available for research and machine learning experimentation. Specifically, the dataset was accessed through the Kaggle notebook titled "Chronic Kidney Disease which is available at: https://www.kaggle.com/code/niteshyadav3103/chronic-kidney-disease. This dataset is originally derived from the chronic kidney disease dataset from the UCI Machine Learning Repository, which is widely used in medical prediction research. The dataset contains 400 patient records with 25 attributes, where 24 attributes represent clinical measurements and the final attribute indicates whether the patient has chronic kidney disease. The features include important medical indicators such as age, blood pressure, specific gravity, albumin level, blood glucose, serum creatinine, hemoglobin level, packed cell volume, and other laboratory test results. These attributes consist of both numerical and categorical data types that reflect the physiological condition of the patient. Because of its rich clinical information and balanced representation of CKD and non-CKD cases, this dataset has been widely used to develop predictive models and analyses the importance of medical features in kidney disease detection. In this study, the dataset is used to perform feature selection using multiple approaches in order to identify the most influential attributes for predicting chronic kidney disease.

2) Data Loading and Environment Setup: The implementation begins by importing the necessary Python libraries required for data analysis, machine learning, and feature selection. Libraries such as `pandas` and `NumPy` are used for handling tabular data and performing numerical computations. Modules from `Scikit-learn` are imported to support tasks such as dataset splitting, feature scaling, feature selection, and model evaluation. Specifically, functions like `train_test_split` help divide the dataset into training and testing subsets. The code is executed in a Google Collab environment, where the Google Drive is mounted using the `drive.mount()` function. This step allows the program to access the dataset stored in the user's Google Drive directory. After mounting the drive, the CKD dataset is loaded into a Pandas Data Frame using the `read_csv()` function. This dataset contains multiple clinical and laboratory attributes

TABLE II: Clinical Attributes Used in Dataset

S.No	Feature	Description	Type	Medical Significance
1	sc	Serum Creatinine	Numerical	Indicates kidney filtration efficiency
2	bu	Blood Urea	Numerical	Shows waste accumulation in blood
3	hemo	Hemoglobin	Numerical	Detects CKD-related anemia
4	pcv	Packed Cell Volume	Numerical	Measures red blood cell concentration
5	bgr	Blood Glucose Random	Numerical	Identifies diabetes-related kidney risk
6	sg	Specific Gravity	Encoded Nominal	Evaluates urine concentration ability
7	al	Albumin	Encoded Nominal	Detects protein leakage in urine
8	dm	Diabetes Mellitus	Encoded Nominal	CKD risk indicator
9	htn	Hypertension	Encoded Nominal	Major CKD risk factor
10	rbc	Red Blood Cells	Encoded Nominal	Indicates kidney tissue condition
11	pc	Pus Cells	Encoded Nominal	Shows urinary infection presence
12	sod	Sodium	Numerical	Maintains electrolyte balance
13	pot	Potassium	Numerical	Detects advanced kidney malfunction
14	wc	White Blood Cell Count	Numerical	Indicates infection or inflammation
15	rc	Red Blood Cell Count	Numerical	Detects CKD-related anemia
16	classification	CKD Status	Nominal	Target output variable

associated with kidney health, along with a target column that indicates whether the patient has chronic kidney disease.

3) Data Pre-processing and Cleaning: Before applying machine learning algorithms, the dataset undergoes several preprocessing steps. First, the column names of the dataset are standardized by removing extra spaces, converting all characters to lowercase, and replacing spaces with underscores. This ensures consistency and prevents errors when referencing column names during analysis. Next, the target variable representing the disease classification is identified and stored separately. Since many medical datasets contain categorical variables such as ckd/not ckd, normal/abnormal indicators, or textual attributes, the code converts all object-type columns into numerical form using the Label Encoder method. Label encoding assigns a unique numeric value to each categorical category, making the data compatible with machine learning algorithms that require numerical inputs. After encoding categorical variables, the dataset is divided into two components: input features (X) and target labels (y). The target column representing CKD classification is removed from the feature matrix so that the model can learn patterns from the remaining clinical attributes.

4) Feature Scaling and Normalization: Machine learning algorithms often perform better when features are scaled to a similar range. Therefore, the code applies to normalize the dataset. Standardization transforms each feature so that it has a mean of zero and a standard deviation of one. This step ensures that attributes with larger numerical ranges do not dominate the learning process. The scaled features are then converted back into a Data Frame structure to retain the original column names for easier interpretation. The first few rows of the transformed dataset are displayed to verify that the scaling process has been applied correctly.

5) Handling Missing Values: Medical datasets frequently contain incomplete records due to missing clinical measurements or unavailable laboratory tests. To address this issue, the program applies mean imputation using the Simple Imputer class. This method replaces missing values in each feature with the mean value of that feature across the dataset. Mean imputation helps preserve the overall statistical distribution of the data while ensuring that machine learning models can process the dataset without errors caused by missing entries. The imputed dataset is stored in a new Data Frame called X imputed, which serves as the cleaned feature set used for subsequent feature selection and model evaluation.

6) Filter-Based Feature Selection: The first feature selection technique implemented in this study is the Filter Method, which evaluates the statistical relationship between each feature and the target variable independently. In this implementation, the ANOVA F-test (f classif) is used to measure how strongly each feature is associated with the CKD classification label. The Select Best function ranks all features based on their statistical scores and selects the top twelve most relevant features. These selected attributes are stored in the variable filter features. The purpose of this step is to identify features that have the strongest individual correlation with CKD diagnosis, without considering interactions between features.

7) Wrapper-Based Feature Selection: The second approach used in the analysis is the Wrapper Method, which evaluates feature subsets based on the performance of a machine learning model. In this code, Recursive Feature Elimination (RFE) is applied with a Logistic Regression classifier as the underlying model. RFE works by repeatedly training the model, ranking features according to their importance, and removing the least significant features in each iteration. This process continues until a predefined number of features remains. In this implementation, RFE selects the top ten features that contribute most effectively to classification accuracy. These selected features are stored in the variable wrapper features.

Unlike filter methods, wrapper methods consider feature interactions and the predictive performance of the model.

8) Embedded Feature Selection: The third feature selection technique employed in the code is the Embedded Method, which integrates feature selection directly into the model training process. A Random Forest classifier is used to determine feature importance. Random Forest is an ensemble learning algorithm that constructs multiple decision trees and aggregates their predictions. During training, the model calculates the importance of each feature based on how effectively it reduces classification error across the trees. After training the model with 100 trees, the feature importance scores are extracted and sorted in descending order. The top nine features with the highest importance scores are selected and stored as embedded features. This approach provides a model-driven perspective on which attributes are most relevant for predicting CKD.

9) Intersection of Selected Features: After obtaining feature subsets from the filter, wrapper, and embedded methods, the code computes the intersection of these feature sets. This step identifies features that are consistently selected by all three methods. Features that appear in all subsets are considered highly reliable predictors of chronic kidney disease because they are supported by multiple independent selection techniques. These common features are stored in the variable final features. The intersection approach helps improve the robustness and interpretability of the feature selection process.

10) Model Evaluation Function: To assess the effectiveness of each feature subset, the code defines a function called evaluate subset. This function first divides the dataset into training and testing sets, using 70 % of the data for training and 30% for testing. A Logistic Regression model is then trained on the selected features. After training, the model predicts the class labels of the test data. The performance of the model is evaluated using four standard classification metrics: Accuracy, Precision, Recall, and F1-Score. Accuracy measures the proportion of correct predictions, precision measures the reliability of positive predictions, recall evaluates the model's ability to detect actual CKD cases, and the F1-score represents the harmonic mean of precision and recall. These metrics provide a comprehensive assessment of the model's classification performance.

11) Evaluation of Feature Subsets: The program then evaluates the predictive performance of each feature selection method separately. Four datasets are created based on the features selected by the Filter, Wrapper, Embedded, and Final Intersection methods. Each subset is passed to the evaluation function, and the resulting performance metrics are stored in a dictionary called results. The results for each method are printed in a structured format, allowing direct comparison of how different feature selection techniques affect the classification performance.

12) Saving Experimental Results: Finally, the selected feature subsets and evaluation results are saved for documentation and further analysis. A `DataFrame` containing the features selected by each method is created and exported to a CSV file named `selected_features.csv`. Similarly, the evaluation metrics for each method are stored in another CSV file called `model_results.csv`. If there are thefeatures, a final dataset containfinal _intersection _dataset.csv. Saving these files ensures that the results can be easily used for reporting, visualization, or further experimentation.

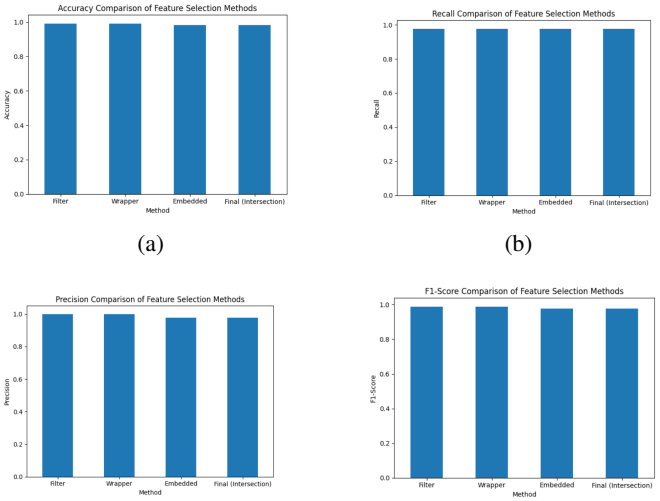


Fig. 2: Evaluation of Feature Selection Methods Across Multiple Performance Metrics

B. Analysis of the Proposed Model Using the CKD Dataset

The proposed model analyzes the CKD dataset using a CNN-based approach to identify important medical features and accurately classify patients as CKD or non-CKD. This dataset contains clinical information collected from patient records and includes a combination of laboratory test results, physiological measurements, and diagnostic indicators that are commonly used by healthcare professionals to assess kidney function. The dataset contains approximately 400 patient instances and 25 attributes, where 24 attributes represent clinical measurements and the final attribute represents the classification label indicating whether the patient suffers from chronic kidney disease. Because the dataset includes a mixture of categorical and numerical attributes such as blood pressure, albumin level, blood glucose level, serum creatinine, hemoglobin, and packed cell volume, it provides a comprehensive representation of the patient's health condition. These attributes allow the proposed CNN-based model to learn meaningful patterns and relationships between clinical variables that contribute to the diagnosis of CKD. The proposed system begins by preparing the dataset for analysis through a series of preprocessing operations. Initially, the dataset is loaded into the computational environment using data handling libraries that enable structured data processing. The column names are standardized by removing unnecessary spaces and converting them into a consistent format to ensure that the dataset is easier to manipulate during analysis. Since the CKD dataset contains categorical variables such as "ckd", "notckd", "present", or "notckd", these values are

converted into numerical representations using label encoding. This transformation allows machine learning algorithms and neural networks to process the information effectively because neural networks require numerical inputs for training. After encoding the categorical attributes, the dataset is separated into two major components: the feature matrix and the target variable. The feature matrix contains all medical attributes used as input variables, while the target variable represents the classification label indicating the presence or absence of chronic kidney disease. Once the dataset is separated into features and target variables, the input features are normalized using a standardization technique. Feature scaling is an essential preprocessing step in neural network-based models because it ensures that all attributes have similar numerical ranges. Without normalization, features with larger numerical values may dominate the learning process and negatively influence model performance. Standardization transforms the dataset so that each feature has a mean value of zero and a standard deviation of one. This process stabilizes the learning behavior of the neural network and improves convergence during training. Another important preprocessing step in the proposed model involves handling missing values. Medical datasets often contain incomplete records due to unavailable laboratory results or missing patient information. To address this issue, missing values in the dataset are replaced using mean imputation. In this approach, the missing entries in each feature are replaced with the average value of that feature calculated from the available data. This strategy ensures that the dataset remains complete while preserving the statistical characteristics of the original data distribution. After the imputation process, the dataset becomes suitable for training the proposed CNN-based classification model. Following the preprocessing stage, the proposed system performs feature selection to identify the most significant attributes that influence the prediction of chronic kidney disease. Feature selection plays an important role in medical data analysis because not all clinical measurements contribute equally to disease prediction. Reducing irrelevant or redundant attributes improves model efficiency and enhances predictive performance. In this research, multiple feature selection approaches are applied to analyze the importance of different clinical variables. The first approach is the filter-based feature selection method, which evaluates the statistical relationship between each feature and the target variable. In this method, the ANOVA F-test is applied to measure the degree of dependency between the medical attributes and CKD classification. The algorithm assigns a score to each feature based on its statistical relevance, and the top features with the highest scores are selected. This process identifies attributes that show strong correlations with the disease outcome. The second approach is the wrapper-based feature selection method, which evaluates subsets of features using a predictive model. In this study, Recursive Feature Elimination (RFE) is applied in combination with logistic regression to determine the most important features. The algorithm repeatedly trains the model while removing the least important feature at each iteration. Through this iterative process, the method gradually identifies the subset of features that contributes most effectively to the prediction task. The third approach is the embedded feature selection method, which integrates feature selection directly within the model training process. A Random Forest classifier is used in this stage to evaluate feature importance based on decision tree structures. Random Forest calculates the contribution of each feature by analyzing how frequently it is used to split nodes within the trees and how much it reduces prediction error. Features with higher importance scores are considered more influential in predicting CKD. The top features identified through this method represent attributes that strongly affect the classification outcome. After obtaining feature subsets from the filter, wrapper, and embedded methods, the proposed model determines the intersection of these feature sets. The intersection process identifies features that appear consistently across all selection methods. These features are considered highly reliable because they demonstrate strong relevance across different analytical approaches. The intersection features represent the final optimized feature set that is used as input for the proposed CNN-based classification model. The CNN model analyzes the selected features to learn complex patterns present in the CKD dataset. Although CNN architectures are commonly associated with image processing, they are also effective in analyzing structured data by capturing local patterns and relationships among features. In this study, the clinical attributes are represented as a one-dimensional input vector, allowing the convolutional layers to detect meaningful correlations between neighboring features. The convolutional filters scan the input feature space and extract significant patterns related to kidney disease indicators. These extracted patterns are then passed through nonlinear activation functions, which introduce nonlinearity into the model and enable it to learn complex relationships between medical attributes. Pooling layers are applied to reduce the dimensionality of the extracted feature maps while retaining the most informative patterns. This process helps reduce computational complexity and prevents overfitting. The resulting feature maps are then flattened and passed through fully connected layers that perform the final classification task. The output layer generates predictions indicating whether a patient is likely to have chronic kidney disease or not. To evaluate the effectiveness of the proposed model, the dataset is divided into training and testing subsets using a standard data splitting technique. Typically, 70 % of the dataset is used for training the model, while the remaining 30% is used for testing. The training data allows the CNN model to learn patterns from the dataset, while the testing data evaluates the model's ability to generalize to unseen patient records. The performance of the model is measured using several evaluation metrics, including accuracy, precision, recall, and F1-score. Accuracy measures the overall correctness of the predictions, while precision evaluates how many predicted CKD cases are actually correct. Recall measures the model's ability to correctly identify patients who truly have CKD. The F1-score combines precision and recall into a single metric, providing a balanced evaluation of model

performance. The analysis of the proposed CNN-based model demonstrates that combining feature selection techniques with deep learning improves the prediction capability of the system. Feature selection reduces noise in the dataset by focusing on the most informative medical attributes, while the CNN architecture effectively captures nonlinear relationships between these attributes. As a result, the proposed model provides a robust framework for early detection and analysis of chronic kidney disease using machine learning techniques.

C. Visualization of the Proposed Model Using a Confusion Matrix

CKD prediction model is evaluated and visualized using a confusion matrix. A confusion matrix provides a clear and structured way to understand how well the classification model distinguishes between CKD and non-CKD cases. It presents the comparison between the actual class labels and the predicted class labels produced by the model. This visualization helps researchers observe the strengths and weaknesses of the classification system and understand how accurately the model identifies patients with chronic kidney disease. The confusion matrix is commonly used in binary classification problems because it summarizes prediction outcomes in a simple tabular form. In the context of the CKD dataset, the confusion matrix consists of four important components: True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN). Each of these components represents a different type of prediction made by the model. True Positive refers to the cases where the model correctly predicts that a patient has CKD. True Negative represents the cases where the model correctly identifies that a patient does not have CKD. False Positive occurs when the model predicts that a patient has CKD even though the patient is actually healthy. False Negative represents the cases where the model fails to detect CKD in a patient who actually has the disease. These four values together provide a complete picture of the model's classification performance. The proposed model uses logistic regression as the classification algorithm after applying feature selection techniques such as filter, wrapper, and embedded methods. Once the model is trained on the selected features, it predicts the CKD status for the test dataset. The predicted results are then compared with the actual class labels, and the confusion matrix is generated to visualize the classification outcomes. This matrix makes it possible to quickly identify how many cases are correctly classified and how many cases are misclassified by the model. Visualizing the confusion matrix helps highlight the reliability of the proposed CKD prediction system. When the values of True Positives and True Negatives are high, it indicates that the model is successfully distinguishing between CKD and non-CKD patients. On the other hand, high values of False Positives or False Negatives suggest that the model may require further optimization. In medical diagnosis, minimizing False Negatives is particularly important because failing to detect a disease in a patient can delay treatment and worsen the patient's health condition. Therefore, the confusion matrix plays a critical role in analyzing the effectiveness of the predictive model in real healthcare scenarios. In addition to providing a visual summary of classification results, the confusion matrix also supports the calculation of several evaluation metrics used in this study. Metrics such as accuracy, precision, recall, and F1-score are directly derived from the values present in the confusion matrix. Accuracy measures the overall correctness of the model's predictions. Precision evaluates the proportion of predicted CKD cases that are actually correct. Recall measures the ability of the model to detect true CKD cases among all actual CKD patients. The F1-score combines precision and recall into a single metric that reflects the balance between them. By analyzing these metrics along with the confusion matrix visualization, the overall performance of the model can be thoroughly assessed. Another important advantage of using the confusion matrix is that it provides an intuitive visual interpretation of the model's predictions. Instead of relying only on numerical performance scores, researchers can directly observe how the predictions are distributed across the four categories of the matrix. This visual representation helps identify whether the model tends to misclassify a particular class more frequently. For example, if the matrix shows a higher number of False Positives, it indicates that the model may incorrectly classify healthy patients as CKD patients. Similarly, a high number of False Negatives suggests that the model is missing actual CKD cases, which can be critical in medical applications. The confusion matrix visualization therefore serves as an essential tool in evaluating the reliability and clinical usefulness of the proposed CKD prediction model. By presenting a clear overview of correct and incorrect predictions, it allows researchers to understand the strengths and limitations of the model. This analysis supports the development of more accurate diagnostic systems that can assist healthcare professionals in early detection and monitoring of CKD.

D. Comparison of Accuracy, Precision, Recall, and F1-Score

The performance of the proposed model is evaluated using four important classification metrics: accuracy, precision, recall, and F1-score. These evaluation measures provide a comprehensive understanding of how effectively the model predicts CKD using the selected features from the dataset. Since medical prediction systems require both reliability and accuracy, relying on a single evaluation metric is not sufficient. Therefore, multiple performance metrics are used to assess different aspects of the model's prediction capability and to compare the effectiveness of different feature selection methods used in the study. Accuracy measures the overall correctness of the model by calculating the proportion of correctly predicted instances out of the total number of observations. It represents how frequently the model correctly classifies patients into CKD and non-CKD categories. In the proposed model, accuracy indicates how well the logistic regression classifier performs after applying the selected feature subsets obtained through filter, wrapper, and embedded feature selection techniques. A higher accuracy value suggests that the model effectively

distinguishes between patients with kidney disease and those without it. However, accuracy alone does not always provide a complete picture of model performance, particularly in medical datasets where the cost of misclassification can be high. For example, a model may achieve high accuracy but still fail to detect several patients who actually have CKD. Because of this limitation, additional evaluation metrics such as precision and recall are also considered. Precision evaluates the correctness of the positive predictions made by the model. It measures the proportion of patients who are predicted as having CKD and actually suffer from the disease. In other words, precision indicates how reliable the positive predictions of the model are. A high precision value means that when the model predicts CKD, it is usually correct. This reduces the number of false positive cases, where a healthy patient is incorrectly classified as having kidney disease. Minimizing false positives is important because incorrect diagnosis can lead to unnecessary medical tests, additional costs, and psychological stress for patients. Therefore, precision plays an important role in assessing how trustworthy the model's positive predictions are. Recall, also known as sensitivity, measures the model's ability to correctly identify patients who truly have chronic kidney disease. It calculates the proportion of actual CKD cases that the model successfully detects. In medical diagnosis systems, recall is extremely important because missing a true CKD case may delay treatment and allow the disease to progress further. A higher recall value indicates that the model effectively identifies most of the patients who are suffering from kidney disease. In other words, it minimizes false negative cases where a patient with CKD is incorrectly predicted as healthy. Since early detection of CKD is crucial for timely medical intervention, recall becomes a critical performance indicator for the proposed model. The F1-score provides a balanced evaluation of the model by combining both precision and recall into a single metric. It is calculated as the harmonic mean of precision and recall, which ensures that both measures contribute equally to the final evaluation. The F1-score becomes particularly useful when there is a need to maintain a balance between correctly identifying disease cases and minimizing incorrect predictions. A high F1-score indicates that the model achieves both high precision and high recall, meaning that it accurately detects CKD patients while also minimizing false alarms. In medical prediction tasks such as CKD diagnosis, maintaining this balance is essential because both types of errors can have significant consequences. In this research, accuracy, precision, recall, and F1-score are used to compare the performance of different feature selection methods applied to the CKD dataset. The filter method selects features based on statistical relationships with the target variable, the wrapper method selects features by evaluating their contribution to model performance, and the embedded method identifies important features during the model training process. Each selected feature subset is used to train the logistic regression classifier, and the resulting predictions are evaluated using the four performance metrics. By comparing these metrics across different feature selection techniques, the study identifies which subset of features provides the most effective representation of the CKD dataset. The evaluation helps determine whether statistical feature selection, model-based feature selection, or a combination of methods produces better predictive performance. Furthermore, the intersection of features selected by multiple methods is also evaluated to identify the most consistent and reliable predictors of chronic kidney disease. Overall, the comparison of accuracy, precision, recall, and F1-score provides a comprehensive assessment of the proposed model. These metrics help analyze the strengths and limitations of the model while also highlighting the importance of selecting relevant medical features for CKD prediction. Through this evaluation, the study demonstrates how feature selection techniques can improve model performance and support the development of more accurate and reliable early detection systems for CKD.

E. Conclusion

While a Hybrid CNN-KNN Model has shown to yield good performance in early prediction of CKD, further improvements on this system may also add value in the analysis of future works. First we aimed to improve the newly developed GCAM-R by including a more heterogeneous and extensive data set that is obtained from different geographically distributed locations of multiple hospitals, which will provide increased generalisation ability for new clinical practices. Further studies can further integrate different patient information, such as lifestyle choices, family history of health, historical records concerning disease and Time Series Data to predict the speed at which CKD develops and make more accurate predictions of CKD stage in advance symptoms occur. Additional models, such as Attention Models and Transformer Networks have Advanced Deep Learning Architectures that can help synthesize features and predict CKD more accurately than what we achieved to this point. Another project opens the opportunity to validate it into clinical setting using HIS and EMR systems where doctors will be able to take real-time decision on factual patient's data. Mobile/web-based apps may also be developed for remote monitoring of patient health to support rural and resource-poor regions. Key future research should be on interpretable AI, in order to make the outputs given by automated predictions more transparent and trustworthy. To facilitate the introduction of intelligent CKD prediction models into routine clinical use, we need to work on interpretable models, validate results using in vivo data from trials and cope with challenges such as privacy concerns or ethical requirements, among others. CKD has emerged as a major challenge for health systems around the world and this can be related to its progressive nature, which increases the burden of treatment, as well as the fact that it is diagnosed mainly in the late stages. The failure to show symptoms during the asymptomatic phase represents a driving force in the death of this disease and its subsequent severe complications. This study was conducted to generate optimal computational models for CKD prediction. In the current investigation, a comprehensive ML model has been

proposed as a diagnosis tool. A robust preprocessing method was followed to address several inherent problems in a health care dataset, which is missing data points, outliers detection and removal, feature scaling and handling of an imbalance class. Inorder to establish the benchmarks a range of SVM, LR, KNN and SGD algorithms were used for the train test split. In the experiment, we found that the standalone model could achieve good prediction accuracy as shown in Table 2 for SVM, and KNN. However, with regard to the efficiency in computations there are differences between the models. In terms of efficiency, the SVM and LR are chosen to be integrated in an ensemble classifier. The purpose has to make them easier to work together and design the voids of each one. Compared with all single classifiers, the presented ensemble method showed better diagnosis ability for CKD prediction by achieving the best result of overall accuracy as well as better performance on every evaluation index such as Precision, Recall, F1-score, ROC curve analysis and Cohen's kappa value. The experiment results demonstrate that the ensemble learning proposed in this work can bring substantial improvement to CKD prediction task in terms of generalization ability and classification error decrease. In this work, a demonstration has also been presented tothe effectiveness of hybrid learning models by using deep neural network-based feature extraction for CKD classification. Based on total results, itis apparent that the use of machine learning based ensemble and hybrid techniques have potential ability for developing trustworthy tools in detection of early stage CKD. It is to be concluded that the proposed method has thepotential to help health care experts for identifying high risk patients at early stage of the disease and then timely management action can be implemented towards patient care. This work may be extended in future using bigger and varied datasets, such as population specific data and advanced deep learning architectures and feature selection methods. These extensions couldfurther enhance diagnostic performance and promote wider clinical use.

REFERENCES

REFERENCES

[1] V. Jha, G. Garcia-Garcia, K. Iseki, Z. Li, S. Naicker, B. Plattner, R. Saran, A. Y.-M. Wang, and C.-W. Yang, "Chronic kidney disease: global dimension and perspectives," *The Lancet*, vol. 382, no. 9888, pp. 260–272, 2013.

[2] E. Dritsas and M. Trigka, "Machine learning techniques for chronic kidney disease risk prediction," *Big Data and Cognitive Computing*, vol. 6, no. 3, p. 98, 2022.

[3] A. S. Levey and J. Coresh, "Chronic kidney disease," *The Lancet*, vol. 379, no. 9811, pp. 165–180, 2012.

[4] N. R. Hill, S. T. Fatoba, J. L. Oke, J. A. Hirst, C. A. O'Callaghan, D. S. Lasserson, and F. D. R. Hobbs, "Global prevalence of chronic kidney disease: A systematic review and meta-analysis," *PLOS ONE*, vol. 11, no. 7, p. e0158765, 2016.

[5] Z. Obermeyer and E. J. Emanuel, "Predicting the future: Big data, machine learning, and clinical medicine," *New England Journal of Medicine*, vol. 375, no. 13, pp. 1216–1219, 2016.

[6] K. Kumar, M. Pradeepa, M. Mahdal, S. Verma, M. V. L. N. RajaRao, and J. V. N. Ramesh, "A deep learning approach for kidney disease recognition and prediction through image processing," *Applied Sciences*, vol. 13, no. 6, p. 3621, 2023.

[7] J. Ma, J. Wang, L. Lu, Y. Sun, M. Feng, F. Zhang, P. Shen, Z. Jiang, S. Hong, and L. Zhang, "Development and validation of a dynamic kidney failure prediction model based on deep learning: A real-world study with external validation," *arXiv preprint arXiv:2501.16388*, 2025.

[8] P. Chittora, S. Chaurasia, P. Chakrabarti, G. Kumawat, T. Chakrabarti, Z. Leonowicz, M. Jasiński, Ł. Jasiński, R. Gono, E. Jasińska, *et al.*, "Prediction of chronic kidney disease—A machine learning perspective," *IEEE Access*, vol. 9, pp. 17312–17334, 2021.

[9] S. Balakrishnan, "Feature selection using improved teaching learning based algorithm on chronic kidney disease dataset," *Procedia Computer Science*, vol. 171, pp. 1660–1669, 2020.

[10] P. Yildirim, "Chronic kidney disease prediction on imbalanced data by multilayer perceptron," in *Proc. 2017 IEEE 41st Annual Computer Software and Applications Conference (COMPSAC)*, vol. 2, pp. 193–198, 2017.

[11] D. A. Debal and T. M. Sitote, "Chronic kidney disease prediction using machine learning techniques," *Journal of Big Data*, vol. 9, no. 1, pp. 1–19, 2022.

[12] M. A. Islam, M. Z. H. Majumder, and M. A. Hussein, "Chronic kidney disease prediction based on machine learning algorithms," *Journal of Pathology Informatics*, vol. 14, p. 100189, 2023.

[13] Y. Zhu, D. Bi, M. Saunders, and Y. Ji, "Prediction of chronic kidney disease progression using recurrent neural network and electronic health records," *Scientific Reports*, vol. 13, no. 1, p. 22091, 2023.

[14] D. Saif, A. M. Sarhan, and N. M. Elshennawy, "Deep-kidney: An effective deep learning framework for chronic kidney disease prediction," *Health Information Science and Systems*, vol. 12, no. 1, p. 3, 2023.

[15] X. Zou, Z. Xie, H. Huang, Y. Lin, R. Chen, Z. Lin, J. Liang, and Y. Zhang, "A multimodal stage prediction study of chronic kidney disease based on deep learning," in *Proc. 2025 2nd International Conference on Electronic Engineering and Information Systems (EEISS)*, pp. 1–6, 2025.

[16] Z. A. Miller and K. Dwyer, "Artificial intelligence to predict chronic kidney disease progression to kidney failure: A narrative review," *Nephrology*, vol. 30, no. 1, p. e14424, 2025.